

COL 703/7203

03 August

$$\frac{\phi \wedge \neg q \rightarrow r, \neg r, \phi}{q}$$

1. $\phi \wedge \neg q \rightarrow r$ premise
2. $\neg r$ premise
3. ϕ premise

4.	$\neg q$	assumption
5.	$\phi \wedge \neg q$	$\wedge i$ 3,4
6.	r	MP 5,1
7.	$\neg r$	copy 2
	$\perp$	$\neg e$ 6,2
	$\neg \neg q$	$\neg i$ 7
	q	$\neg \neg e$

Derived rules NP

$$\frac{\phi \quad \phi \rightarrow \psi}{\psi}$$

$\phi \rightarrow \psi$

$\neg \psi$

ϕ

ψ

$\perp$

$\neg \phi$

premise

premise

assumption

MP

$\neg_e(\psi, \neg \psi)$

MT

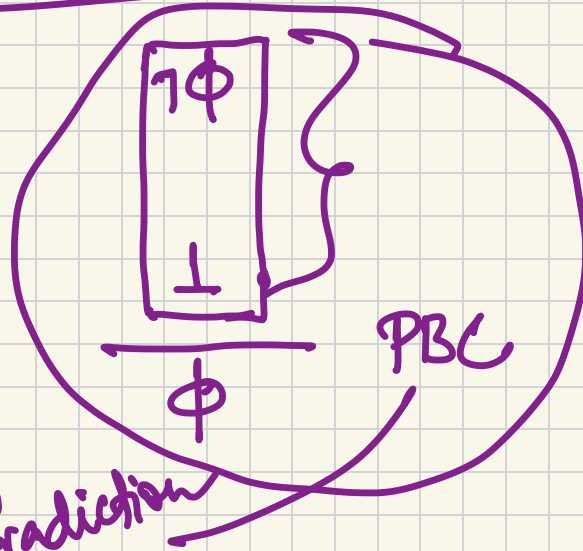
$$\boxed{\frac{\phi \rightarrow \psi \quad \neg \psi}{\neg \phi}} \quad \text{Derived rule}$$

$\neg\neg i$ is also a derived rule.

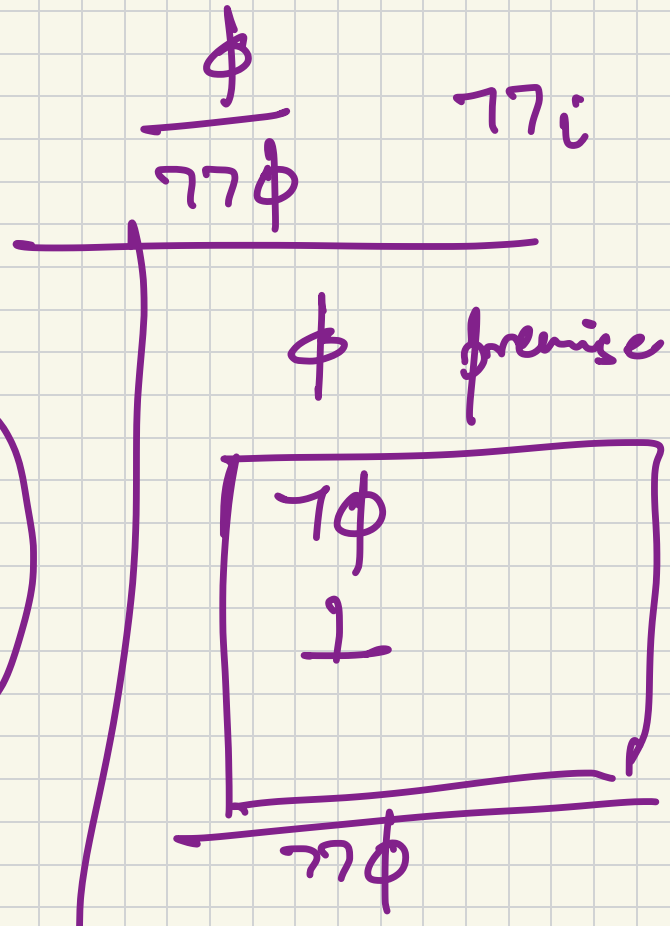
Two more derived rules

$$\frac{\neg\phi \rightarrow \perp}{\phi}$$

1.



Proof by contradiction



2. Law of excluded middle

$$\overline{\phi \vee \neg \phi} \quad \text{LEM}$$

1.	$\neg(\phi \vee \neg \phi)$	assumption
2.	ϕ	assumption
3.	$\phi \vee \neg \phi$	$\vee i_1$
4.	$\perp$	$\neg e \ 1, 3$
5.	$\neg \phi$	$\neg i \ 2-4$
6.	$\phi \vee \neg \phi$	$\vee i_2$
7.	$\perp$	$\neg e \ 1, 6$

1 PBC

Use LEM to prove that

$\begin{array}{l} p \rightarrow q \\ p \rightarrow q \\ p \vee \neg p \end{array}$	$\vdash \neg p \vee q$
premise	premise
LEM	LEM
$\begin{array}{l} p \text{ assum} \\ q \text{ MP} \\ \neg p \vee q \end{array}$	$\begin{array}{l} \neg p \\ \neg p \vee q \end{array}$

Exercise

$$p \rightarrow q \quad \vdash \quad \neg p \vee q$$

1. $p \rightarrow q$ premise
2. $q \vee \neg q$ LEM

$$\vdash \rightarrow \quad \neg \vdash \quad \neg \phi \vee q$$

provable equivalence

$$\phi_1 \neg \vdash \phi_2 \equiv \vdash (\phi_1 \rightarrow \phi_2) \wedge (\phi_2 \rightarrow \phi_1)$$

$$\underline{(\phi \vee \neg \phi)}$$

There exists irrational nos. a and b
such that a^b is rational.

$$\begin{array}{l} (\sqrt{2})^{\sqrt{2}} \\ \downarrow \\ \text{rational} \end{array} \quad \begin{array}{l} \downarrow \\ \text{irrational} \\ \hline (\sqrt{2})^{\sqrt{2}} \end{array}$$

$$\begin{array}{l} \sqrt{2} \text{ is irrational} \\ \hline \sqrt{2} = p/q \end{array}$$

Propositional logic as a formal language

$$(p \rightarrow q) \wedge r$$

$$pq \rightarrow \wedge r \vdash \wedge pr$$

Infinitely many atoms / atomic propositions

$$\{ \wedge, \vee, \neg, \rightarrow, (,) \}$$

Well-formed formulas (wff)

- every atomic proposition is a wff.
- if ϕ is a wff, $(\neg \phi)$ is a wff.
- if ϕ and ψ are wffs,
 $(\phi \vee \psi)$
 $(\phi \wedge \psi)$
 $(\phi \rightarrow \psi)$ are wffs.
- nothing else is a wff

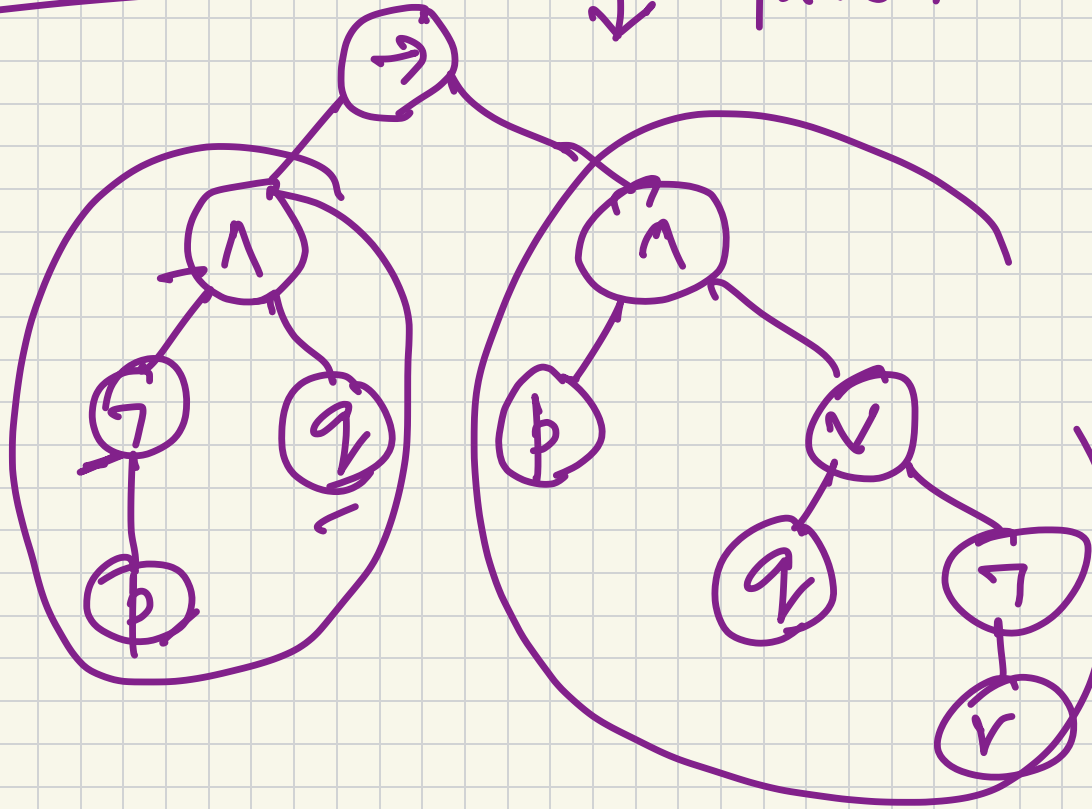
$$\boxed{(((\neg p) \wedge q) \rightarrow (p \wedge (q \vee (\neg r))))}$$

Subformulas

p
 q
 r
 $\neg p$
 $\neg r$
 $\neg p \wedge q$
 $q \vee \neg r$

$p \wedge (q \vee \neg r)$

Parse tree



BNF (Backus Naur Form)

$$\phi ::= p \mid (\neg \phi) \mid (\phi_1 \rightarrow \phi_2) \mid (\phi_1 \vee \phi_2) \mid (\phi_1 \wedge \phi_2)$$

Semantics of propositional logic

$$\phi_1, \phi_2, \dots, \phi_n \vdash \psi$$

$$\underbrace{\phi_1}, \underbrace{\phi_2}, \dots, \underbrace{\phi_n} \models \psi$$



semantic entailment
(entails)

Atomic propositions are
declarative sentences

↗ which get one of the
truth values

T
(true)

(if they match
the reality of
our world)

F
(false)

(if they do
not match
the reality
of our world)

Meaning of the logical symbols

ϕ
T
T
F
F

ψ
T
F
T
F

$\phi \wedge \psi$
T
F
F
F

$\phi \vee \psi$
T
T
T
F

$\phi \rightarrow \psi$
T
F
T
T

ϕ	$\neg \phi$
T	F
F	T

Valuation



an assignment of truth
values to the
atomic propositions